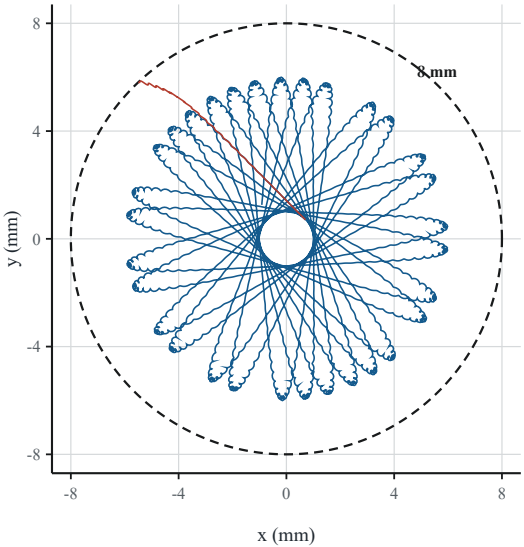
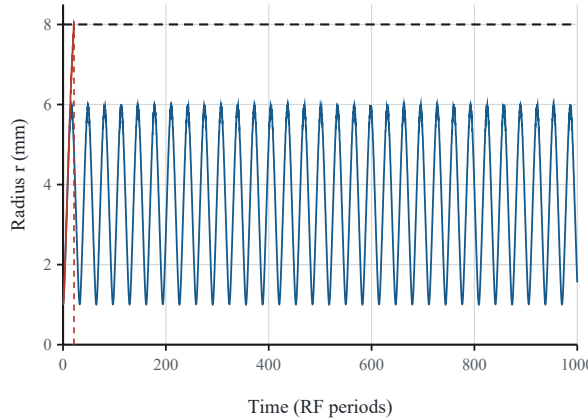
Matched launch: rotating retained, standing escaped

m/z = 100 kTh · E_perp = 1 eV · r0 = 1 mm · θ = 45° · φ_RF = 0° · + tangential · 250 V each

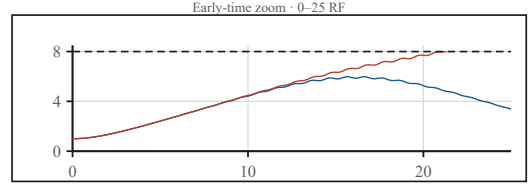
(a) x–y trajectory (mm)



(b) Radius versus RF period



Early-time zoom · 0–25 RF



Rotating · retained to 1000 RF

Standing · escaped at 21.31 RF

Dashed circle/line: 8-mm aperture. Saved traces use 8 samples per RF; the short panel expands the escape interval.